FOUNDATION COURSE

START YOUR PREPARATION FROM BASICS

Prerequisites of Engineering Mathematics

Lecture no. 09

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Recap *of previous lecture*

(BINOMIAL THEOREM)



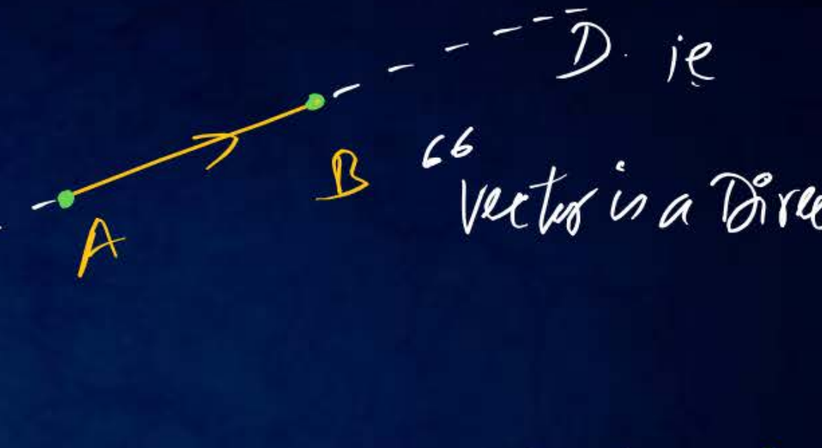
Topics to be covered

"VECTORS"



Defⁿ: Vector Quantity consist following three properties;

1. Magnitude = $|\vec{AB}|$
2. Direction / Sense $A \to B$
3. Line of Support / Line of Action (CD)



ie "Vector is a Directed Line Segment"

Types: (1) Localized Vectors → these consist above 3 properties for eg force

(2) Free Vectors → these vectors consist only 1st two properties. Vectors used in Maths are Free Vectors.

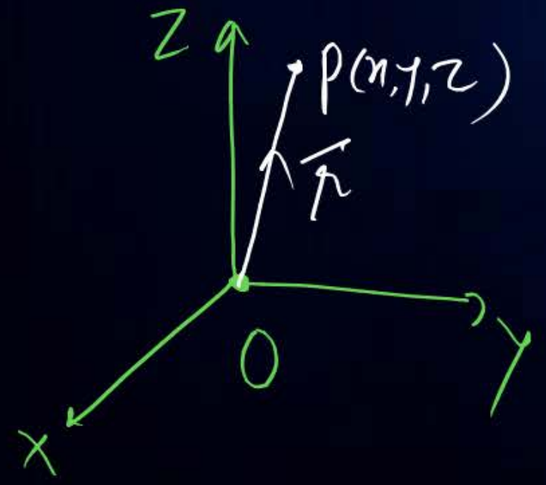
(*) Free Vectors can be moved anywhere in space.

Zero vector / Point vector – A vector whose magnitude is zero is called zero vector, & it's dir^n is **Not defined** (bcoz any line can be taken as it's dir^n)

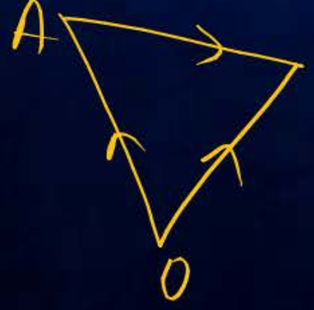
Generally origin is considered as Point vector i.e $\vec{O} = 0\hat{i}+0\hat{j}+0\hat{k} \simeq (0,0,0)$

Position vector of any Point $P(x,y,z)$ w.r. to origin → it is given as;

$$\vec{OP} = x\hat{i}+y\hat{j}+z\hat{k} \quad \text{i.e } \vec{r} = \vec{OP} = x\hat{i}+y\hat{j}+z\hat{k} \simeq (x,y,z)$$



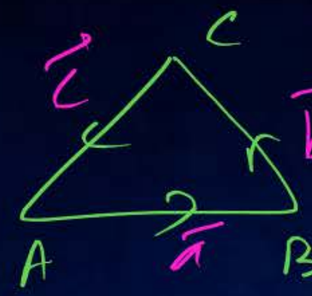
Note ① Let $A(x_1,y_1,z_1)$ & $B(x_2,y_2,z_2)$ ⟨ $\vec{OA} = x_1\hat{i}+y_1\hat{j}+z_1\hat{k}$, $\vec{OB} = x_2\hat{i}+y_2\hat{j}+z_2\hat{k}$



By △ law of Vectors; $\vec{OA}+\vec{AB} = \vec{OB}$

i.e $\boxed{\vec{AB} = \vec{OB}-\vec{OA}} = (x_2-x_1)\hat{i}+(y_2-y_1)\hat{j}+(z_2-z_1)\hat{k}$

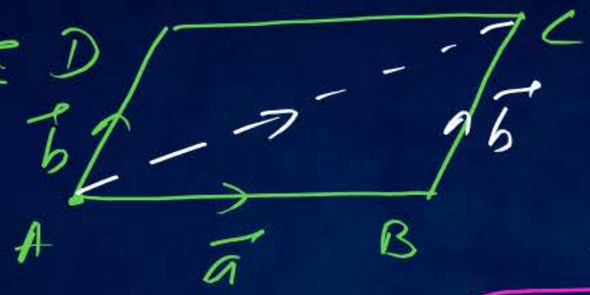
(2) Δ Law of vectors :-



$\vec{AB}+\vec{BC}+\vec{CA}=\vec{0}$

ie $\vec{a}+\vec{b}+\vec{c}=\vec{0}$.

$\parallel^{gm}$ Law of vectors :-



$\vec{AB}+\vec{AD}=\vec{AC}$

ie $\vec{AC}=\vec{a}+\vec{b}$.

(3) Let $\vec{a}=a_1\hat{i}+a_2\hat{j}+a_3\hat{k}$ then $|\vec{a}|=\sqrt{a_1^2+a_2^2+a_3^2}$

& unit vector $\hat{a}=\dfrac{\vec{a}}{|\vec{a}|}$ $\Rightarrow \vec{a}=|\vec{a}|\,\hat{a}$

& $\vec{a}\cdot\vec{a}=|\vec{a}|^2=a_1^2+a_2^2+a_3^2$

Any Vector = (Magnitude) (unit Vector in it's dir^n)

eg: unit vector along x axis $= \hat{i}$ & vector of magnitude 5 units along x axis $= 5\hat{i}$

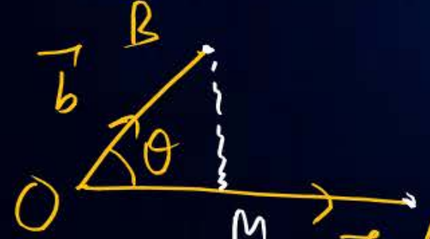
" " " y " $= \hat{j}$ " " " 3 " " y axis $= 3\hat{j}$

" " " z " $= \hat{k}$ " " " 7 " " z axis $= 7\hat{k}$

Dot Product (Scalar Product) → Let $\vec{a}$ & $\vec{b}$ are two given vectors then

$$\boxed{\vec{a}\cdot\vec{b} = |\vec{a}||\vec{b}|\cos\theta} = |\vec{a}|(|\vec{b}|\cos\theta) = |\vec{a}|(OM) = \begin{cases} \text{stretch} : OM > 1 \\ \text{Shrink} : 0 < OM < 1 \end{cases}$$

= Scalar Quantity.



$$\frac{OM}{OB} = \cos\theta \Rightarrow OM = |\vec{b}|\cos\theta$$

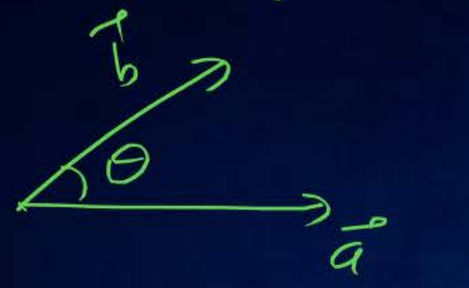
Note (1) if $\vec{a}\cdot\vec{b} = 0$ then both the vectors are $\perp^r$.

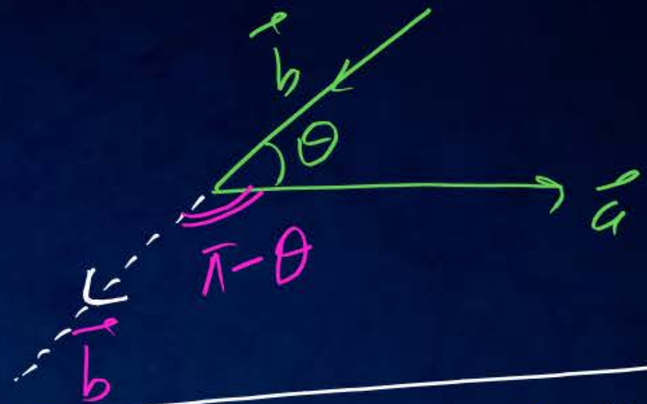
i.e Condition for perpendicularity is $\vec{a}\cdot\vec{b} = 0$

(2) With the help of above Concept, we can find angle b/n two vectors ie $\theta = \cos^{-1}\left[\frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|}\right]$

③ Angle b/n two vectors is defined only when either both are converging or both are diverging.

$0 \le \text{Angle} \le \pi$





& Angle = $\pi - \theta$

& Angle b/n $\vec{a}$ & $\vec{b}$ = θ

Cross product → it is defined as: $\boxed{\vec{a} \times \vec{b} = (|\vec{a}|\,|\vec{b}|\sin\theta)\,\hat{n}}$ = vector Quantity

= (Magnitude)(unit vector in $\perp^r$ dir^n)

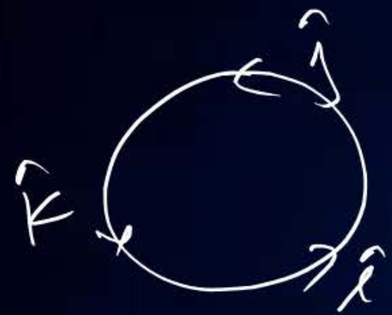
ie $|\vec{a} \times \vec{b}| = |\vec{a}|\,|\vec{b}|\sin\theta$ & $\hat{n} = \dfrac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|}$ = unit vector $\perp^r$ to the plane containing $\vec{a}$ & $\vec{b}$

& $\vec{a} \times \vec{b}$ = $\perp^r$ vector to the plane containing $\vec{a}$ & $\vec{b}$

= Normal vector

= unit Normal vector.

① Condition for $\parallel^r$ vectors is $\boxed{\vec{a}\times\vec{b}=\vec{0}}$ i.e $\hat{i}\times\hat{i}=\hat{j}\times\hat{j}=\hat{k}\times\hat{k}=\vec{0}$

② 

$\hat{i}\times\hat{j}=\hat{k}$ But $\hat{j}\times\hat{i}=-\hat{k}$, $\hat{i}\times\hat{k}=-\hat{j}$ & so on.....

$\hat{j}\times\hat{k}=\hat{i}$ & $\hat{k}\times\hat{j}=-\hat{i}$

③ $\boxed{\vec{a}\cdot\vec{b}=|\vec{a}|\,|\vec{b}|\cos\theta}\Rightarrow \hat{i}\cdot\hat{j}=\hat{j}\cdot\hat{k}=\hat{k}\cdot\hat{i}=0$ & $\hat{i}\cdot\hat{i}=\hat{j}\cdot\hat{j}=\hat{k}\cdot\hat{k}=1$

Condition for $\perp^r$ vectors is $\boxed{\vec{a}\cdot\vec{b}=0}$

④ Lagrange's Identity :-

$$\boxed{|\vec{a}\times\vec{b}|^2+(\vec{a}\cdot\vec{b})^2=a^2b^2}$$

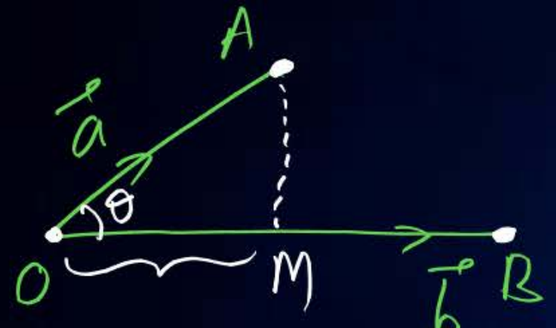
where $a=|\vec{a}|$ & $b=|\vec{b}|$

Proof: W.K.T $\cos\theta=\dfrac{\vec{a}\cdot\vec{b}}{ab}$ & $\sin\theta=\dfrac{|\vec{a}\times\vec{b}|}{ab}$

Now $\sin^2\theta+\cos^2\theta=1$

$$\frac{|\vec{a}\times\vec{b}|^2}{a^2b^2}+\frac{(\vec{a}\cdot\vec{b})^2}{a^2b^2}=1\Rightarrow|\vec{a}\times\vec{b}|^2+(\vec{a}\cdot\vec{b})^2=a^2b^2$$

(5) Component of $\vec{a}$ in the dir^n of $\vec{b}$ → it is given as OM & is defined as $\vec{a}\cdot\hat{b}$



$$\frac{OM}{OA}=\cos\theta \Rightarrow OM = OA\cos\theta$$

$$= |\vec{a}|\cos\theta = |\vec{a}|\left(\frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|}\right) = \frac{\vec{a}\cdot\vec{b}}{|\vec{b}|} = \vec{a}\cdot\hat{b}$$ //

(6) $\boxed{\vec{a}\cdot\vec{b}=\vec{b}\cdot\vec{a}}$ But $\boxed{\vec{a}\times\vec{b}=-(\vec{b}\times\vec{a})}$

(7) Scalar Tripple Product (S.T.P) → it is defined as $\boxed{[\vec{a}\,\vec{b}\,\vec{c}]=\vec{a}\cdot(\vec{b}\times\vec{c})}$

Note (i) it is the representation of Volume of Cuboid with edges $\vec{a},\vec{b},\vec{c}$ | = Scalar Quantity = Volume of Cuboid

(2) Condition for Coplanar vectors is $\boxed{[\vec{a}\,\vec{b}\,\vec{c}]=0}$ i.e. STP = 0

(9) Vector Triple Product → it is defined as $\boxed{\vec{a}\times(\vec{b}\times\vec{c})=(\vec{a}\cdot\vec{c})\vec{b}-(\vec{a}\cdot\vec{b})\vec{c}}$

eg Evaluate $\vec{a}\times(\vec{b}\times\vec{c})+\vec{b}\times(\vec{c}\times\vec{a})+\vec{c}\times(\vec{a}\times\vec{b})=?$

$\vec{a}\times(\vec{b}\times\vec{c})=(\vec{a}\cdot\vec{c})\vec{b}-(\vec{a}\cdot\vec{b})\vec{c}$ ——— (1)

$\vec{b}\times(\vec{c}\times\vec{a})=(\vec{b}\cdot\vec{a})\vec{c}-(\vec{b}\cdot\vec{c})\vec{a}$ ——— (2)

$\vec{c}\times(\vec{a}\times\vec{b})=(\vec{c}\cdot\vec{b})\vec{a}-(\vec{c}\cdot\vec{a})\vec{b}$ ——— (3)

$\therefore\ \vec{a}\times(\vec{b}\times\vec{c})+\vec{b}\times(\vec{c}\times\vec{a})+\vec{c}\times(\vec{a}\times\vec{b})=\vec{0}$

eg P.T. $\hat{i}\times(\vec{a}\times\hat{i})+\hat{j}\times(\vec{a}\times\hat{j})+\hat{k}\times(\vec{a}\times\hat{k})=2\vec{a}$

Let $\vec{a}=a_1\hat{i}+a_2\hat{j}+a_3\hat{k}$

$\hat{i}\times(\vec{a}\times\hat{i})=(\hat{i}\cdot\hat{i})\vec{a}-(\hat{i}\cdot\vec{a})\hat{i}=1\vec{a}-a_1\hat{i}$

$\hat{j}\times(\vec{a}\times\hat{j})=(\hat{j}\cdot\hat{j})\vec{a}-(\hat{j}\cdot\vec{a})\hat{j}=1\vec{a}-a_2\hat{j}$

$\hat{k}\times(\vec{a}\times\hat{k})=(\hat{k}\cdot\hat{k})\vec{a}-(\hat{k}\cdot\vec{a})\hat{k}=1\cdot\vec{a}-a_3\hat{k}$

Adding LHS $=3\vec{a}-(a_1\hat{i}+a_2\hat{j}+a_3\hat{k})$

$=3\vec{a}-\vec{a}=2\vec{a}=$ RHS

Shortcuts :- Let $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$, $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$

$$\vec{a}\cdot\vec{b} = a_1b_1 + a_2b_2 + a_3b_3$$

$$\vec{a}\times\vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

$$\& \ [\vec{a}\,\vec{b}\,\vec{c}] = \vec{a}\cdot(\vec{b}\times\vec{c}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

Note: $\vec{a}\cdot\vec{a} = a_1a_1 + a_2a_2 + a_3a_3 = a_1^2 + a_2^2 + a_3^2 = |\vec{a}|^2$

Q: Find the angle b/n the vectors $\vec{a}$, $\vec{b}$ s.t they are of same Magnitude ($\sqrt{2}$) & their scalar Product is -1 ?

Sol: Atq, $|\vec{a}| = |\vec{b}| = \sqrt{2}$

$\vec{a}\cdot\vec{b} = -1$

$$\& \ \cos\theta = \frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|} = \frac{-1}{\sqrt{2}\sqrt{2}} = \frac{-1}{2}$$

$\theta = 120^\circ$.

Q. Find λ for which the vectors $\vec{a}=3\hat{i}+2\hat{j}+9\hat{k}$ & $\vec{b}=1\hat{i}+\lambda\hat{j}+3\hat{k}$ are (1) $\perp^r$ (2) $\parallel^r$.

(1) $\because \vec{a}$ & $\vec{b}$ are $\perp^r$ so $\vec{a}\cdot\vec{b}=0$

$(3)(1)+(2)(\lambda)+(9)(3)=0$

$\lambda=-15$

(2) $\because \vec{a}$ & $\vec{b}$ are $\parallel^r$ so $\vec{a}\times\vec{b}=\vec{0}$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 2 & 9 \\ 1 & \lambda & 3 \end{vmatrix}=\vec{0}$$

$\hat{i}(6-9\lambda)-\hat{j}(9-9)+\hat{k}(3\lambda-2)=\vec{0}$

$(6-9\lambda)\hat{i}+0\hat{j}+(3\lambda-2)\hat{k}=0\hat{i}+0\hat{j}+0\hat{k}$

Comparing coeff. of $\hat{i}$ both sides.

$6-9\lambda=0 \Rightarrow \lambda=2/3$

Q. If $|\vec{a}|=10$, $|\vec{b}|=2$, $\vec{a}\cdot\vec{b}=12$ then $|\vec{a}\times\vec{b}|=?$

Using Lagrange's Identity, $|\vec{a}\times\vec{b}|^2+(\vec{a}\cdot\vec{b})^2=a^2b^2$

$|\vec{a}\times\vec{b}|^2+(12)^2=(10)^2(2)^2$

$|\vec{a}\times\vec{b}|^2=400-144=256$

$|\vec{a}\times\vec{b}|=\sqrt{256}=\pm 16$

Q. Find a vector of magnitude 9 units & which is $\perp^r$ to both the vectors $4\hat{i}-\hat{j}+3\hat{k}$ & $-2\hat{i}+\hat{j}-2\hat{k}$

Sol: $\vec{a}\times\vec{b}$ = $\perp^r$ vector to the plane containing $\vec{a}$ & $\vec{b}$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & -1 & 3 \\ -2 & 1 & -2 \end{vmatrix}$$

$$= \hat{i}(2-3) - \hat{j}(-8+6) + \hat{k}(4-2)$$

$\vec{a}\times\vec{b} = -\hat{i}+2\hat{j}+2\hat{k}$ = Normal Vector

$$|\vec{a}\times\vec{b}| = \sqrt{(-1)^2+(2)^2+(2)^2} = 3$$

So unit vector $\perp^r$ to the plane of $\vec{a}$ & $\vec{b}$

$$\hat{n} = \frac{\vec{a}\times\vec{b}}{|\vec{a}\times\vec{b}|} = \frac{-\hat{i}+2\hat{j}+2\hat{k}}{3}$$

So Req. vector = (It's Magnitude)(unit vector)

$$= 9\,\hat{n}$$

$$= 9\left(\frac{-\hat{i}+2\hat{j}+2\hat{k}}{3}\right)$$

$$= -3\hat{i}+6\hat{j}+6\hat{k}$$

Q Find the Projection of $\vec{a} = 7\hat{i}+\hat{j}-4\hat{k}$ on $\vec{b} = 2\hat{i}+6\hat{j}+3\hat{k}$

= Component

Sol: $\vec{a} = 7\hat{i}+1.\hat{j}-4\hat{k}$

$|\vec{b}| = \sqrt{4+36+9} = 7$

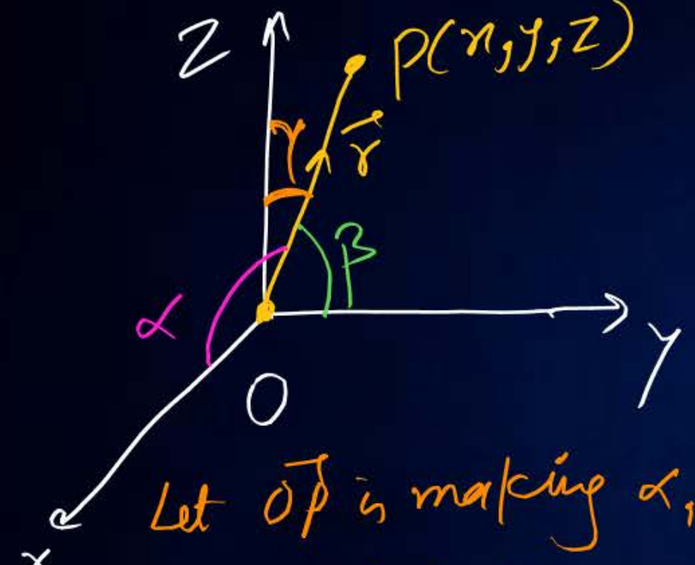
$\hat{b} = \frac{\vec{b}}{|\vec{b}|} = \left(\frac{2}{7}\right)\hat{i}+\left(\frac{6}{7}\right)\hat{j}+\left(\frac{3}{7}\right)\hat{k}$

So Req. Projection $= a.\hat{b}$

$= (7)\left(\frac{2}{7}\right)+(1)\left(\frac{6}{7}\right)+(-4)\left(\frac{3}{7}\right)$

$= \frac{14+6-12}{7} = \frac{8}{7}$

Direction Cosine & Direction Ratio →



$\vec{r} = \vec{OP}$
$= x\hat{i} + y\hat{j} + z\hat{k}$

Let $\vec{OP}$ is making α, β & γ angle with the dir^n of X, Y & Z axis resp.

then $\langle \cos\alpha, \cos\beta, \cos\gamma \rangle$ is called D. Cosines of $\vec{OP}$. where $\boxed{\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1}$

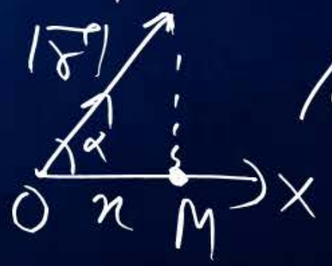
Note: ① Let $l = \cos\alpha$, $m = \cos\beta$, $n = \cos\gamma$

then $\langle l, m, n \rangle$ are the D. Cosines s.t $l^2 + m^2 + n^2 = 1$

② $\vec{r} = \vec{OP} = x\hat{i} + y\hat{j} + z\hat{k}$

$\Rightarrow |\vec{OP}| = |\vec{r}| = r = \sqrt{x^2 + y^2 + z^2}$

③ $\cos\alpha = \frac{OX}{OP} = \frac{x}{|\vec{r}|} \Rightarrow l = \frac{x}{|\vec{r}|} \Rightarrow \boxed{x = l|\vec{r}|}$



Similarly, $\cos\beta = \frac{y}{|\vec{r}|} \Rightarrow m = \frac{y}{|\vec{r}|} \Rightarrow \boxed{y = m|\vec{r}|}$

& $\cos\gamma = \frac{z}{|\vec{r}|} \Rightarrow n = \frac{z}{|\vec{r}|} \Rightarrow \boxed{z = n|\vec{r}|}$

i.e. $x = l|\vec{r}|,\ y = m|\vec{r}|,\ z = n|\vec{r}|$

where $\langle l, m, n \rangle$ are the D. Cosines

& $\langle x, y, z \rangle$ are the D. Ratios

(4) for the vector $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$;

D. Cosines are $\langle l, m, n \rangle$ or $\langle \cos\alpha, \cos\beta, \cos\gamma \rangle$

or $\left\langle \frac{x}{|\vec{r}|}, \frac{y}{|\vec{r}|}, \frac{z}{|\vec{r}|} \right\rangle$

& D. Ratios are $\langle x, y, z \rangle$ or $\langle l|\vec{r}|, m|\vec{r}|, n|\vec{r}| \rangle$

(5) P.T. $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

Proof: $\because \vec{r} = \vec{OP} \Rightarrow r = |\vec{OP}|$

$|\vec{OP}|^2 = r^2$

$x^2 + y^2 + z^2 = r^2$

$(l|\vec{r}|)^2 + (m|\vec{r}|)^2 + (n|\vec{r}|)^2 = r^2$

$l^2r^2 + m^2r^2 + n^2r^2 = r^2$

$l^2 + m^2 + n^2 = 1$

(6) P.T. $\boxed{\sin^2\alpha+\sin^2\beta+\sin^2\gamma=2}$

w.k that $\cos^2\alpha+\cos^2\beta+\cos^2\gamma=1$

$(1-\sin^2\alpha)+(1-\sin^2\beta)+(1-\sin^2\gamma)=1$

$3-(\sin^2\alpha+\sin^2\beta+\sin^2\gamma)=1$

i.e $\sin^2\alpha+\sin^2\beta+\sin^2\gamma=2$

(*) Representation of vector in terms of Direction

Let $\vec{r}=x\hat{i}+y\hat{j}+z\hat{k}$ is the given vector.

$=(l|\vec{r}|)\hat{i}+(m|\vec{r}|)\hat{j}+(n|\vec{r}|)\hat{k}$

$\boxed{\vec{r}=|\vec{r}|[l\hat{i}+m\hat{j}+m\hat{k}]}$

Any Vector = (it's Magnitude) (unit vector in the dir^n of $\vec{r}$)

$\vec{r}=|\vec{r}|\,\hat{r}$ where $\hat{r}=l\hat{i}+m\hat{j}+n\hat{k}$

$|\vec{r}|=\sqrt{l^2+m^2+n^2}=1$

i.e unit vector in the dir^n of $\vec{r}=\hat{r}=l\hat{i}+m\hat{j}+n\hat{k}$

$=(\cos\alpha)\hat{i}+(\cos\beta)\hat{j}+(\cos\gamma)\hat{k}$

Q: ① D. Cosines of x axis = ? = $\langle 1,0,0 \rangle$

② " " " y axis = ? = $\langle 0,1,0 \rangle$

③ " " " z axis = ? = $\langle 0,0,1 \rangle$

Sol ① For x axis; $\alpha = 0, \beta = \frac{\pi}{2}, \gamma = \frac{\pi}{2}$

So D Cosines are $\langle \cos\alpha, \cos\beta, \cos\gamma \rangle$

$= \langle \cos 0, \cos \pi/2, \cos \pi/2 \rangle$

$= \langle 1,0,0 \rangle$

Q: A vector $\vec{OP}$ is inclined with x & y axis by 45° & 60° then find the Angle which $\vec{OP}$ makes with Z axis?

Sol: $\alpha = 45^\circ, \beta = 60^\circ, \gamma = ?$

w.k.that $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

$\cos^2\frac{\pi}{4} + \cos^2\frac{\pi}{3} + \cos^2\gamma = 1$

$\frac{1}{2} + \frac{1}{4} + \cos^2\gamma = 1$

$\cos^2\gamma = 1 - \frac{3}{4} = \frac{1}{4} \Rightarrow \cos\gamma = \pm\frac{1}{2}$

$\Rightarrow \gamma = 60^\circ$ or 120°

Q: A vector is equally inclined with coordinate axes then find it's D. Cosines.

Sol: ΔABC; $\alpha = \beta = \gamma$

$\Rightarrow \cos\alpha = \cos\beta = \cos\gamma \Rightarrow \boxed{l = m = n}$

Now; $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

$l^2 + m^2 + n^2 = 1$

$l^2 + l^2 + l^2 = 1 \Rightarrow l^2 = \frac{1}{3}$

$l = \pm\frac{1}{\sqrt{3}} = m = n$

Soo D Cosines are $\left\langle \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right\rangle$ or $\left\langle \frac{-1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}, \frac{-1}{\sqrt{3}} \right\rangle$

(ii) If magnitude of above vector is 6 units then find the vector.

Sol: $\vec{r} = |\vec{r}|\left(l\hat{i} + m\hat{j} + n\hat{k}\right)$

$= 6\left[\pm\frac{1}{\sqrt{3}}\hat{i} \pm \frac{1}{\sqrt{3}}\hat{j} \pm \frac{1}{\sqrt{3}}\hat{k}\right]$

$= \pm 2\sqrt{3}\left(\hat{i} + \hat{j} + \hat{k}\right)$ =

eg: if D.Cosines are $\left\langle \frac{2}{3}, \frac{-2}{3}, \frac{1}{3} \right\rangle$

then D.Ratios are ?

Ans: DR are $\langle 2, -2, 1 \rangle$

Ans " " $\langle -2, 2, -1 \rangle$ & so on

⊛ if $\langle a, b, c \rangle$ are the D.Ratios

then D.Cosines are

$$\left\langle \frac{\pm a}{\sqrt{a^2+b^2+c^2}}, \frac{\pm b}{\sqrt{a^2+b^2+c^2}}, \frac{\pm c}{\sqrt{a^2+b^2+c^2}} \right\rangle$$

eg: if D.Ratios are $\langle 3, -4, 12 \rangle$ then find it's D.Cosines ?

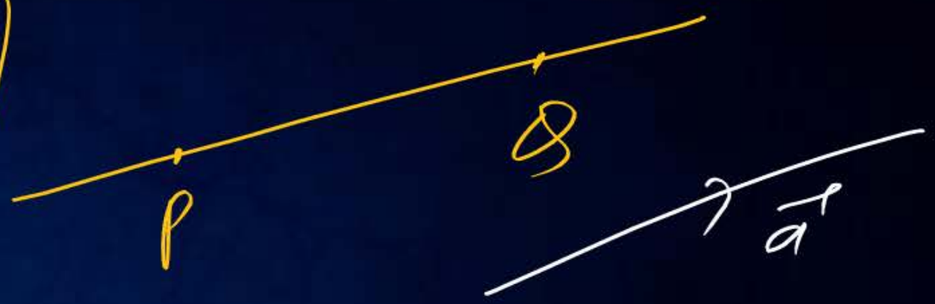
Sol: $\sqrt{a^2+b^2+c^2} = \sqrt{9+16+144} = 13$

So D.Cosines are

$$\left\langle \frac{3}{13}, \frac{-4}{13}, \frac{12}{13} \right\rangle$$

eqn of line in 3D → passing through $P(x_1, y_1, z_1)$ & $Q(x_2, y_2, z_2)$ is;

PQ: $$\boxed{\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1}}$$



D. Ratio of line PQ are; $\langle (x_2-x_1), (y_2-y_1), (z_2-z_1) \rangle$

↳ this line is ∥ to vector $\vec{a} = (x_2-x_1)\hat{i} + (y_2-y_1)\hat{j} + (z_2-z_1)\hat{k}$

& D. Cosines of PQ are $\left\langle \frac{(x_2-x_1)}{|\overline{PQ}|}, \frac{(y_2-y_1)}{|\overline{PQ}|}, \frac{(z_2-z_1)}{|\overline{PQ}|} \right\rangle$


Thank
YOU